00 — Serverless Compute on GCP: Overview

Exam Relevance

Section 3.3 — Deploying applications to serverless compute platforms (GCP Associate Cloud Engineer). You must know:

- Which serverless options Google Cloud offers and their differences
- When to choose each (Cloud Run vs. Cloud Functions vs. App Engine)
- How event-driven architectures fit on top of these platforms (Pub/Sub, Cloud Storage events, Eventarc)

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1. The Three Pillars of GCP Serverless Compute

Service	Unit of Deployment	Best For
Cloud Run	Container image	HTTP/gRPC microservices, web apps, APIs, event consumers (any language)
Cloud Functions	Source code (single function)	Lightweight event handlers, glue code, simple APIs
App Engine	Source code (whole app)	Traditional web applications with platform-managed runtime

All three:

- Auto-scale from zero (App Engine Standard and Cloud Run)
- Have built-in HTTPS endpoints
- Integrate with IAM, Cloud Logging, Cloud Monitoring, Cloud Trace
- Support custom domains
- Can be triggered by HTTP requests **or** Google Cloud events (via Eventarc, Pub/Sub, or direct triggers)

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2. Quick Decision Tree

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Do you have a container?
├── YES → Cloud Run
└── NO
    ├── Single function reacting to events? → Cloud Functions
    ├── Whole web app, multiple endpoints? → App Engine
    └── Need full container control + scale-to-zero? → Build a container,
use Cloud Run
```

> **Modern recommendation (2024+):** Google now positions **Cloud Run** as the default serverless platform. **Cloud Functions 2nd gen is built on Cloud Run**, so there is increasing convergence.

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3. Comparison Matrix (Exam-Critical)

Feature	Cloud Run	Cloud Functions (Gen 2)	App Engine Standard	App Engine Flex
Input	Container image	Source code	Source code	Source code or container
Languages	Any (container)	Node, Python, Go, Java, .NET, Ruby, PHP	Node, Python, Go, Java, PHP, Ruby	Any (Docker)
Cold start	Yes (mitigated by min-instances)	Yes	Very fast	Slower (VM-based)
Scale to zero	✅ Yes	✅ Yes	✅ Yes	❌ Min 1 instance
Max request time	60 min	60 min (HTTP) / 9 min (event)	10 min	60 min
Stateful	No	No	No	No
Concurrency / instance	Up to 1000	Up to 1000 (Gen 2)	1 (per dynamic instance)	Configurable
VPC access	Direct VPC or Connector	Direct VPC or Connector	Connector	Native VPC
Pricing	Per request + CPU/mem (per 100ms)	Per invocation + CPU/mem	Instance-hours	Instance-hours
Free tier	2M requests/month	2M invocations/month	28 instance-hours/day	None

4. Event-Driven Architecture on GCP Serverless

Three primary ways to trigger serverless workloads from Google Cloud events:

a) Pub/Sub Push Subscriptions

- Pub/Sub message → HTTPS push to your Cloud Run / Cloud Functions / App Engine endpoint
- Simple, direct, reliable, with retry built-in

b) Eventarc (the modern, unified router)

- Routes events from **150+ Google Cloud sources** (Cloud Storage, BigQuery, Cloud SQL, Audit Logs, etc.) to Cloud Run, Cloud Functions, GKE, or Workflows
- Uses CloudEvents specification
- Auto-creates Pub/Sub topics and subscriptions under the hood
- **Recommended approach for new event-driven apps**

c) Cloud Storage Direct Triggers

- Cloud Functions can be directly triggered by GCS object events (legacy 1st gen, or via Eventarc in 2nd gen)
- Gen 2 functions and Cloud Run use **Eventarc** under the hood for GCS events

```
┌──────────────┐
│ Event Source │   (GCS, Pub/Sub, Audit Log, BigQuery, Firestore, ...)
└──────┬───────┘
       │
       ▼
┌──────────────┐
│   Eventarc   │   ← unified router (CloudEvents)
└──────┬───────┘
       │
       ▼
┌─────────────────────────────────────┐
│  Cloud Run / Cloud Functions / GKE  │
└─────────────────────────────────────┘
```

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5. What This Series Covers

File	Topic
`01-cloud-run.md`	Deploy, configure, manage Cloud Run services
`02-cloud-functions.md`	Cloud Functions 1st gen vs. 2nd gen, deployment, runtimes
`03-app-engine.md`	App Engine Standard and Flexible environments
`04-pubsub-event-processing.md`	Triggering serverless from Pub/Sub
`05-cloud-storage-events.md`	Reacting to Cloud Storage object change notifications
`06-eventarc.md`	Eventarc — unified event routing
`07-choosing-serverless-service.md`	Decision matrix and trade-offs
`08-exam-tips-practice-scenarios.md`	Exam traps, keywords, scenario walkthroughs

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6. Exam-Relevant Notes

- **"Container, scale to zero" → Cloud Run.**
- **"Single function, react to event" → Cloud Functions.**
- **"Existing app, just upload code" → App Engine Standard.**
- **"Route an event from any GCP service to compute" → Eventarc.**
- App Engine **Flex does not scale to zero** — minimum 1 instance.
- Cloud Functions Gen 2 is built on **Cloud Run** + **Eventarc** internally.
- Pub/Sub **push subscriptions** are the simplest way to fan out events to HTTP endpoints.
- For **GCS object events** in Gen 2 / Cloud Run → use **Eventarc** trigger.

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7. Sources

- Cloud Run docs
- Cloud Functions docs
- App Engine docs
- Eventarc docs
- Choosing a serverless option